Automated Load Sharing in Power Distribution Systems: A Transformer-Based Approach Using Arduino Microcontrollers

Sujit Kumar
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
sujitaryan0412@mail.com

Amit Kumar
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
amitkrvineet246@gmail.com

Kriti kumari
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
kritikumari073@gmail.com

Kumari Beauty Rani
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
kumaribeautyrani519@gmail.com

Alka Kumari
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
alkakri110@gmail.com

Abhinav Kumar
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
abhinavfes35@gmail.com

Ankit Kumar
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
ankitkr980164@gmail.com

Varsha kumari
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
vashanavi123456@gmail.com

Md. Zafar
Electrical Engineering Department
Government Engineering College Banka
Bihar, India
mdzafarsabour35@gmail.com

***Abstract*—Transformers are critical components in electrical power distribution systems, yet they are prone to inefficiencies and potential damage under overload conditions. This paper presents the development of an automated load-sharing system to address these challenges, employing Arduino microcontrollers for dynamic load management between primary and backup transformers. The proposed system integrates sensors and a switching mechanism to detect abnormal load conditions. When an overload is identified, the Arduino activates the backup transformer, redistributing the load to prevent thermal stress and maintain continuous power supply. Additionally, the system allows for seamless alternation between transformers during normal operating conditions, ensuring uninterrupted power delivery and facilitating maintenance activities. The implementation of this system has demonstrated significant advantages, including enhanced reliability of the power distribution network, reduced operational risks, and prolonged transformer lifespan. The integration of automation and efficient load-sharing techniques has optimized transformer performance while minimizing the likelihood of equipment failure.**

***Index Terms*—Sensors, Microcontroller, Arduino and Automatic Load Sharing**

I. Introduction

In modern power distribution systems, transformers play a vital role in ensuring reliable power delivery to consumers. However, the demand for electricity is highly variable, resulting in significant fluctuations in transformer loads. Transformers are designed to operate within specific load capacities, and any consistent overloading can lead to excessive thermal stress, damaging the windings and potentially causing permanent failure. Given the critical and costly nature of transformers, their protection from overload conditions is imperative to maintain the reliability of power systems.

This paper focuses on the development of an automated load-sharing system for transformers to address the challenges posed by overload conditions. The proposed system leverages microcontrollers and relays to dynamically monitor and manage transformer loads. Under normal operating conditions, the primary transformer handles the load independently. However, when the load exceeds the predefined capacity of the primary transformer, the system automatically activates a secondary transformer in parallel. This dynamic load-sharing approach reduces the burden on the primary transformer, prevents thermal overloading, and ensures uninterrupted power supply.

Manual intervention for load-sharing operations is often unreliable and inefficient, especially under real-time conditions. By automating this process, the proposed system enhances operational reliability and efficiency. Additional features, such as sensors for precise load monitoring and alarms for overload warnings, further improve the system's responsiveness. Moreover, the automated load-sharing mechanism alternates between transformers, preventing excessive thermal stress and facilitating routine maintenance without disrupting the power supply.

The paper also highlights potential advancements in the system, including priority-based load shedding during extreme overload conditions to safeguard critical infrastructure such as hospitals and industries. The implementation of this automated load-sharing system not only ensures system reliability but also minimizes operational risks and prolongs transformer lifespan, demonstrating the necessity and practicality of adopting such an approach in smart power distribution networks.

V.N Chatse et al., [1] had proposed automatic load sharing in the transformer improves efficiency, reliability and safety of the power system by using IoT. reliability, and safety of power systems. It stops overloads and keeps the load balanced, making transformers work better and last longer. With real-time monitoring, temperature control, and automated load management, the power supply stays stable, lowering costs and reducing problems. IoT- based automatic load sharing system help distribute load efficiently, protecting transformer and ensuring a reliable power supply. This method protects the infrastructure and helps energy management grow, benefiting both industries and homes while supporting sustainable national growth. Yogesh K. Kirange et al., [2] had investigated that using an automatic load-sharing system with Arduino UNO makes power distribution better. It helps manage power between main and backup transformers, stopping overloads, reducing risks, and keeping power on. it prevents overloading, minimizes operational risks, and ensures uninterrupted power supply. The system needs less manual work, uses resources better, and makes transformers last longer. In the future, we could include IoT, machine learning for maintenance, and better monitoring to make it even better for today's modern power distribution network.

Shailesh G. Thakre et al., [3] had demonstrated in Load Sharing of Transformer Using Microcontroller that the parallel operation of transformers, managed automatically through a microcontroller, can effectively protect transformers under overload conditions, prevent damage, and ensure uninterrupted power supply. By sharing the load between two transformers, the system increases reliability, avoids overheating, and maintains efficiency. The automation eliminates manual intervention and optimizes transformer operation particularly under increasing load demands. Krushna K. Tormad et al., [4] had proposed in load Sharing of Transformers Automatically by Using Arduino with GSM that this paper shows that the automatic load-sharing system provides uninterrupted power supply and protects transformers from overloading and overheating. It increases the efficiency and reliability of the system while reducing human intervention. The system allows for alternating switching between transformers, enabling natural cooling and ensuring continuous power delivery.

Prafull Pawar et al., [5] investigated the concept of automatic load sharing between transformers using Programmable Logic Controllers (PLC). Their study focused on managing load distribution to prevent damage caused by overloading and overheating. By employing PLC-based control, transformers connected in parallel can automatically distribute the load when one transformer exceeds its capacity. This approach enhances system reliability, reduces operational costs, and ensures an uninterrupted power supply to consumers. Furthermore, the system eliminates the need for manual intervention, minimizes stress on transformers, and significantly extends the lifespan of the equipment. Aniket Kuntawar et al., [6] studied the potential of load sharing between transformers for substation applications. By intelligently sharing the load between transformers, the system avoids unnecessary operation of multiple transformers during off-peak hours, leading to substantial energy savings and reduced operational costs. Furthermore, the automated connection of additional transformers during peak loads ensures reliable power supply and prevents system overload. Vivek Vishnupant Chouguler et al., [7] monitored the parameters and discovered that if the load on one transformer increases, the relay will sense the change in current, the microcontroller will activate, and slave transformers will immediately activate to share the load. They designed and tested the work on 'Automatic load sharing transformers' and fabricated a demo unit for operating three transformers in parallel to automatically share the load with the help of a change over relay and relay driver circuit, as well as to protect the transformers from overloading and thus provide an uninterrupted power supply to the customers. Natumanya Akimu et al., [8] presented a project aimed at monitoring and mitigating overload conditions within an electrical system. The primary objective of the study was to detect overloaded lines using current sensors and subsequently isolate the affected line to ensure the safety and uninterrupted operation of other lines. The system integrates current sensors, Arduino, and relays to automatically disconnect the overloaded line, thereby automating the overload management process. This approach enhances system reliability, ensures the safety of equipment, and maintains continuous electricity supply to unaffected locations.

The main contribution of this work is the development of an innovative automated load-sharing system that effectively addresses the challenges of transformer overloading and power distribution inefficiencies. Employing Arduino microcontrollers, the proposed system facilitates dynamic load management between primary and backup transformers. It integrates advanced sensors and a switching mechanism to detect abnormal load conditions in real-time. Upon identifying an overload, the system activates the backup transformer, ensuring the redistribution of the load to mitigate thermal stress and maintain a continuous power supply. Furthermore, the system enables seamless alternation between transformers during normal operating conditions, promoting uninterrupted power delivery and facilitating routine maintenance activities without compromising system reliability. The implementation of this automated solution has demonstrated substantial benefits, including enhanced reliability of the power distribution network, reduced operational risks, and extended transformer lifespan. By combining automation and efficient load-sharing techniques, the system optimizes transformer performance while minimizing the probability of equipment failure. This contribution underscores the potential of integrating microcontroller-based automation into power distribution networks to achieve sustainable, reliable, and efficient energy infrastructure. The organization of this paper is as follows: Section II Presents the Methodology, Section III presents the Results and Discussion and Section IV summarized the conclusion.

II. Methodology

The proposed system for automatic load sharing among transformers employs an Arduino-driven framework to ensure optimized load distribution and prevent transformer overloading. The methodology focuses on real-time monitoring and dynamic load management to enhance system reliability and operational efficiency. The core components of the system include transformers, current sensors, voltage sensors, relays, capacitors, an LCD display, and the Arduino Uno microcontroller. The system operates by continuously monitoring electrical parameters, including current and voltage, using sensors integrated with each transformer. These sensors provide real-time data to the Arduino Uno, which processes the information and compares it with predefined safety thresholds. In the event of an overload on the primary transformer, the Arduino triggers relay modules to redistribute the excess load to backup transformers. This dynamic switching mechanism ensures load balancing while minimizing thermal stress on individual transformers. Capacitors are incorporated into the system to stabilize the power supply to the components, ensuring consistent performance. An LCD display provides a user-friendly interface, showcasing real-time load conditions and transformer statuses. Additionally, an alarm feature is included to alert operators when transformers approach their capacity limits. A manual override switch is also integrated into the system to allow flexibility and ensure safety during critical operations or maintenance. Fig. 1. shows the Block Diagram of Automatic Load Sharing of Transformer Using ARDUINO

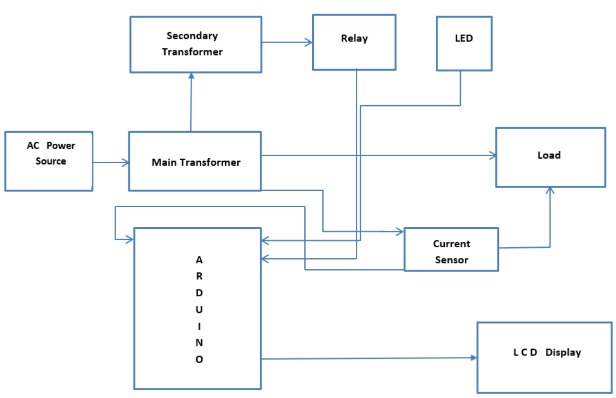

Fig. 1. Block Diagram of Automatic Load Sharing of Transformer Using ARDUINO

The entire setup uses an efficient combination of hardware and software. Current sensors, such as ACS712, measure the load on each transformer, while voltage sensors monitor transformer output. Relays, controlled by the Arduino through programmed logic, enable seamless load transfers. The system is designed to support sustainable and efficient power management by integrating renewable energy sources and accommodating future scalability. This methodology not only ensures continuous power supply but also extends transformer lifespan, reduces operational costs, and enhances overall system reliability, making it suitable for residential, industrial, and rural applications.

A. Connection Diagram

The connection diagram for the Automatic Load Sharing Transformer system is designed to facilitate seamless integration between the components has been shown in Fig. 2. An Arduino Uno microcontroller serves as the central control unit, interfacing with the LCD display, relay module, and current sensor. The LCD display, responsible for showing real-time data, is connected to digital pins 2, 3, 4, 5, 6, and 7 of the Arduino. Specifically, pins 4, 5, 6, and 7 of the Arduino are mapped to the corresponding data pins (D4, D5, D6, and D7) of the LCD, while pin 2 of the Arduino is connected to the enable (E) pin of the LCD display.

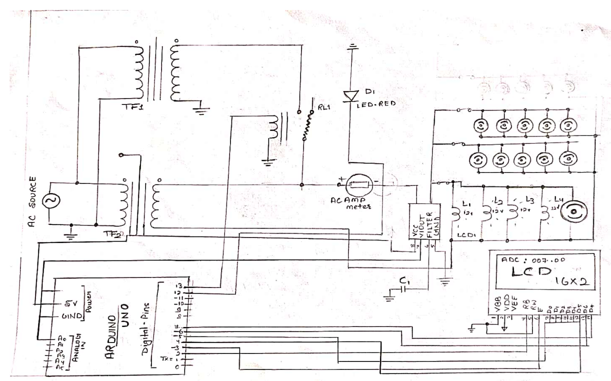

Fig. 2. Connection Diagram

The relay module, which switches the transformers based on load conditions, is connected to digital pin 13 of the Arduino. Additionally, the LED on the relay module, which provides a visual indication of its state, is connected to digital pin 12. This ensures that the system can dynamically activate or deactivate transformers based on the load. The current sensor, used to monitor the electrical load, is connected to the Arduino with three connections: the Vcc pin of the sensor is powered with 5V from the Arduino, the ground (GND) pin is connected to the Arduino's ground, and the output voltage (Vout) pin is linked to the analog input pin (AO) of the Arduino. This setup ensures accurate current measurement and enables the Arduino to make real-time decisions for load management.

This robust connection scheme ensures efficient communication among the components, enabling reliable and automatic operation of the load-sharing system.

The Fig. 3 provides a comprehensive outline of different operating conditions and the corresponding switching mechanisms for two transformers to manage varying load scenarios efficiently. It defines four primary conditions: No Load, Normal Load, Balanced Load, and Overload.

In the No Load condition, there is no electrical demand on the system. Both switches, S1 and S2, are off, meaning both transformers, T1 and T2, are inactive. The LCD display indicates that both transformers are off, and the total load is zero.

During the Normal Load condition, the load remains within the capacity of a single transformer (236.9 Watts). In this scenario, T1 alone handles the entire load, with switch S1 turned on and S2 turned off. The LCD display reflects this state, showing T1 as active and T2 as inactive.

Under the Balanced Load condition, the total load surpasses the capacity of a single transformer, necessitating load sharing. Both transformers, T1 and T2, are activated, with switches S1 and S2 turned on. The load is evenly distributed between the two transformers for improved efficiency, and the display indicates that both are active and sharing the load equally.

Finally, in the Overload condition, the load exceeds the capacity of a single transformer, requiring both transformers to be active. However, T1 handles a larger portion of the load compared to T2. Both switches are on, and the LCD display highlights the system's overload state.

This structured approach to load sharing, as outlined in the lookup table, ensures optimal transformer operation, real-time status visualization, and efficient load management under various system conditions.

S. No.	CONDITION	SWITCH (ON)	DISPLAY			LOAD SHARING (A)		LOAD in Watts		
			T1	T2	Total Load (A)	T1 (A)	T2 (A)	T1	T2	Total Load
01	No load	OFF	OFF	OFF	0	0	0	0	0	0
02	Normal load	S1 ON	ON	OFF	1.03	1.03	0	236.9	0	236.9
03	Balanced load	S2 ON	ON	OFF	2.64	2.64	0	607.2	0	607.2
04	Overload	S3 ON	ON	ON	4.9	2.1	2.1	483	483	966

Fig. 3. Operating Conditions

III. RESULTS AND DISCUSSION

In this work, the simulation was conducted using Proteus software. Fig. 4 to Fig. 7 illustrates the simulation diagram of the automatic load-sharing system for transformers. The simulation demonstrates four distinct modes of operation: No Load, Normal Load, Balanced Load, and Overload. Each mode is explained in detail, accompanied by corresponding simulation diagrams.

Fig. 4 illustrates the No Load Mode, where five bulbs are connected through switches (e.g., S1, S2, S3), and all switches are in the OFF state. As a result, the load current is 0.0 A, and the relay remains in the OFF condition. In this mode, both Transformer 1 and Transformer 2 are inactive, and the LCD display shows a value of 512. Fig. 5 represents the Normal Load Mode, where five bulbs are connected in parallel, with switch S1 in the ON state. The load current in this configuration is 1.03 A, and the relay remains in the OFF condition. Consequently, Transformer 1 becomes active, while Transformer 2 remains inactive, and the LCD display indicates a value of 570. Fig. 6 depicts the Balanced Load Mode, in which five bulbs are connected, and switch S2 is in the ON state. The load current in this mode is 2.64 A, with the relay still in the OFF condition. Under these circumstances, Transformer 1 remains active, and Transformer 2 stays inactive, with the LCD showing a value of 667. Finally, Fig. 7 shows the Overload Mode, where five bulbs are connected, and switch S3 is in the ON state. The load current increases to 4.9 A, causing the relay to switch to the ON condition. In this mode, both Transformer 1 and Transformer 2 are active to share the load, and the LCD display indicates a value of 759.

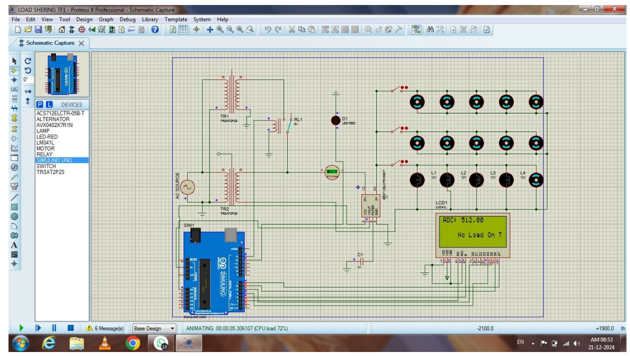

Fig. 4. No Load Mode

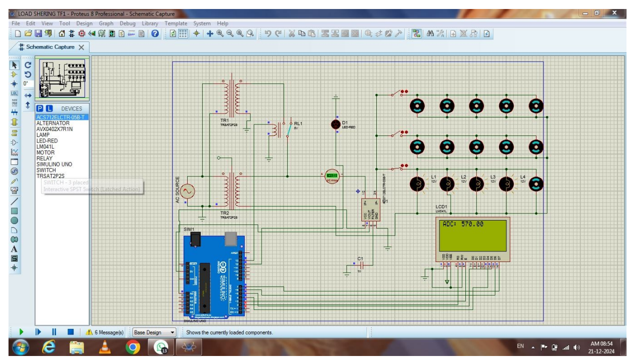

Fig. 5. Normal Load Mode

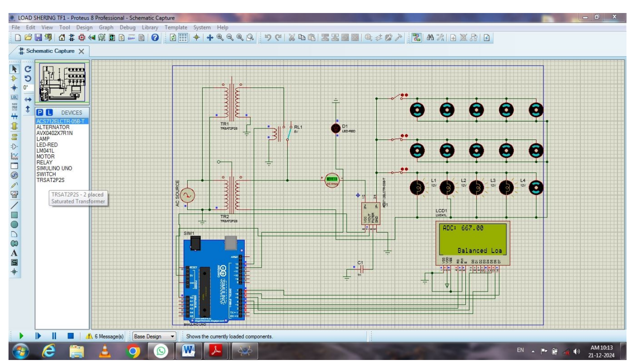

Fig. 6. Balanced Load Mode

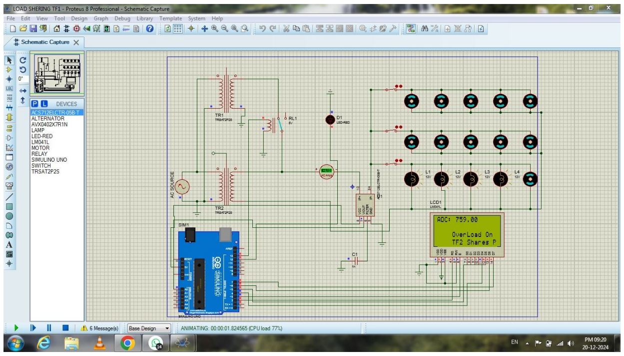

Fig. 7. Overload Mode

IV. CONCLUSION

The development and implementation of the automatic load-sharing transformer system have demonstrated its effectiveness in enhancing the efficiency, reliability, and sustainability of power distribution networks. By ensuring real-time monitoring and optimal load distribution between transformers, the system successfully prevents overloading, mitigates thermal stress, and ensures a smooth and continuous power supply. This not only improves the overall performance of the power system but also extends the lifespan of transformers and reduces operational costs, making it a valuable solution for residential, industrial, and rural applications. The integration of automation and dynamic control into power distribution systems ensures sustainability and resilience in energy infrastructure. The system's ability to efficiently share loads during overload conditions and facilitate maintenance activities without disrupting power supply highlights its practical utility in modern power grids. Several opportunities exist to further enhance the capabilities of automatic load-sharing transformers. Integration with smart grid technologies can enable real-time communication and adaptive control, further improving load balancing and efficiency. Incorporating renewable energy sources will promote sustainable energy use, while modular and scalable designs will enhance flexibility and adaptability to varying power demands. These advancements will position automatic load-sharing transformer systems as a cornerstone in the development of reliable, efficient, and sustainable power grids of the future.

REFERENCES

[1] Pawar, Prafull, Nitin Gite, and Manoj Sonawane, "Automatic Load Sharing of Transformer By using PLC," 2020.

[2] Chouguler, Vivek Vishnupant, and L. S. Patil, "Automatic load sharing of transformer and parameter monitoring," International Journal of Research in Engineering, Science and Management, vol.5, no. 4, pp. 20-21, 2022.

[3] Kirange, Yogesh K., V. S. Patil, Rupesh S. Patil, Rohan Jagdish Patil, and Pawara Dakshata Pandit, "Optimizing power distribution: Smart load sharing mechanism enhanced by arduino uno integration," Int. J. Ingenious Res. Invention Dev, vol. 10, no. 3, pp. 161-174, 2024.

[4] Tormad, Krushna K., Dhiraj A. Nikalje, Ishwar H. Patil, and Dhanashree A. Gayakhe, "Load Sharing of Transformers Automatically by using Arduino with GSM," IC-RTETM-2022 Organized by GGSF's GGS Polytechnic, Nashik, 2022.

[5] V.N Chatse , G.D. Shingade, "Automatic Load Sharing of Distribution Transformer (IOT)". International Journal for Research in Applied Science & Engineering Technology (IJRASET), Volume 12 Issue VI June 2024.

[6] Aniket Kuntawar, Kamlesh Bawane, Ayush Kumar, Swapnil Andi, Lata B. Awale," Automatic Load Sharing of Transformer", International Journal of Research in Engineering, Science and Management, Volume-3, Issue-2, February-2020.

[7] Natumanya Akimu,"Design and Construction of an Automatic Load Monitoring System on a Transformer in Power Distribution Networks", International Digital Organization for Scientific Research ISSN: 2550-794X IDOSR JOURNAL OF SCIENTIFIC RESEARCH 7(1) 58-76, 2022.

[8] Vivek Vishnupant Chouguler, L. S. Patil," Automatic Load Sharing of Transformer and Parameter Monitoring", International Journal of Research in Engineering, Science and Management, vol. 5, no. 4, 2022.